Designed anisotropy on a manufacturability front: fiber-orientation optimization for elastic stiffness, wave control, and phononic band structure

Joshua R. Tempelman

George W.~Woodruff School of Mechanical Engineering Georgia Institute of Technology Atlanta GA

Abstract

Continuous-fiber composites are among the few directly manufacturable materials whose anisotropy can be *steered point by point*, making the local fiber orientation a genuine design field for the material tensor. But a laid tow cannot bend arbitrarily tightly: automated fiber placement (AFP) imposes a minimum turning radius, so the fiber-path curvature is bounded. Performance wants sharp steering; manufacturing forbids it. We show that this single tension organizes fiber-orientation design across *three* regimes into one object — a *manufacturability front* of achievable performance versus allowed tow curvature — and that the manufacturable design is a chosen point on that front, never the unconstrained optimum. (i) *Static stiffness:* on a fixed topology the optimized fibers reach 64% of the compliance benefit of a pointwise-optimal (but unmanufacturable) layup; at the printable end the featured bracket is $9.4\times$ stiffer than an equal-mass isotropic plate at $0.14\times$ its peak stress. (ii) *Wave cloaking*, the sharpest instance: the transformation-elastodynamics cloak sits at the *singular, unreachable* end of the axis (its material diverges, Norris) and, imposed on a real fiber, cloaks only $1.9\times$; free optimization restores the far field $36\times$ but exceeds the print limit; the curvature-constrained optimum is genuinely printable at $9.6\times$. (iii) *Wave focusing:* a lens with $47\times$ focus gain becomes a printable $12\times$ one. (iv) *Periodic media:* fiber orientation on a Bloch cell flattens a chosen band $4.7\times$ and opens complete gaps on both sides. The formulation is deliberately verifiable across all four: a real four-field split of the complex time-harmonic problem, an automatic discrete adjoint, an *exact isotropy-annihilation identity* the orientation gradient must satisfy (a built-in correctness certificate) verified to $\sim 10^{-6}$, and conforming — not soft-density — voids that carry provably zero interior energy.

Keywords: designed anisotropy, fiber-orientation optimization, tow-steered composites, manufacturability, automated fiber placement, elastic wave control, elastic cloaking, phononic band gaps, adjoint sensitivity

1. Introduction

Tailoring how a solid carries load, propagates waves, or filters vibration is, at bottom, tailoring its *anisotropy*. The two dominant ways to engineer anisotropy each pay a manufacturing price. Transformation methods prescribe, exactly, the material tensor an ideal device would need, but that tensor is usually anisotropic beyond any ordinary solid and, for elastic waves, is often *singular*: a transformation cloak requires infinite inertia or an unbounded stiffness ratio at the cloaked surface [2, 3]. Metamaterials realize effective anisotropy with sub-wavelength lattices, inclusions, or local resonators, but the microstructure is intricate and rarely a solid, load-bearing part.

Continuous-fiber-reinforced polymers occupy a different and underused point. A carbon tow has a large but *bounded* stiffness contrast (here $E_{f1}/E_{f2} \approx 14.6$), and AFP lets that anisotropy be *oriented locally*, tow by tow, over a solid part. The orientation field $\theta(\mathbf{x})$ is therefore a directly manufacturable design variable for the material tensor — what transformation methods ask for, minus the singularity. The catch is equally physical: a tow has a minimum turning radius R, so the fiber-path curvature obeys $|\zeta| \leq \zeta_{\text{all}} = 1/R$. *Performance wants sharp steering; the process forbids it.*

Thesis.. This paper argues that the tension above is not an obstacle to be tuned away but the *organizing structure* of fiber-orientation design. For a fixed part it defines a *manufacturability front*: the best achievable performance as a function of the allowed tow curvature ζ_{all}, running from gently-steered, easily-printed designs to a pointwise or transformation ideal that no process reaches. The manufacturable design is a chosen point on that front. We show the front is the same object across three physical regimes governed by the *same* rotated fiber tensor $\mathbf{D}_f(\theta)$ — static stiffness, time-harmonic wave control, and Bloch band structure — and that elastic cloaking is its sharpest instance, because there the ideal endpoint is provably singular and optimization on the front beats the analytical prescription by roughly five to one.

Contributions..

1. **The manufacturability front as a unifying design object**, demonstrated on a static stiffness/strength bracket, a wave cloak and lens, and a phononic Bloch cell — all with one constitutive model, one orientation field, and one curvature constraint (§4,6).
2. **Cloaking as the singular endpoint.** The transformation cloak fails twice for a real composite: its material is singular, so the fiber curvature diverges and it cannot be laid; and imposed on a bounded-contrast fiber it cloaks only $1.9\times$. Free optimization restores the far field $36\times$; the curvature-constrained optimum is printable at $9.6\times$ (§6.2).
3. **One lever, three physics, one certificate.** A single closed-form sensitivity, built entirely from $\partial\mathbf{D}_f/\partial\theta$, serves the harmonic adjoint, the compliance adjoint, and the eigenvalue sensitivity; a trigonometric decomposition of $\mathbf{D}_f(\theta)$ yields an *exact isotropy-annihilation identity* the gradient must satisfy, giving a built-in correctness test (§2,3).
4. **Clean modelling.** A real four-field split of the complex time-harmonic vector problem, and conforming traction-free voids with provably zero interior energy, in place of complex arithmetic and soft-density voids (§2,5).

Relation to prior work.. The manufacturing machinery — a compactly-supported radial-basis (CS-RBF) orientation field and a curvature constraint tied to a minimum turning radius — is that of Wong, Sanders and Rosen [1], whose work designs *static, maximum-stiffness* parts; we do not claim those ingredients. Elastic cloaking is well studied: exact transformation theory and its singular material requirement [2]; manufacturable *analytical* cloaks for *flexural* plate waves by affine transformations [4]; topology-optimized *acoustic* cloaks [5]; and single-phase anisotropic elastic metamaterials by topology optimization [6]. Phononic band gaps are routinely opened by topology or shape optimization of inclusions. Our contribution is neither the fiber machinery nor any one device, but the demonstration that *one* manufacturability front unifies stiffness,

wave, and periodic design in a solid steered-fiber medium, with cloaking as the instance where the ideal is singular and optimization on the front is the only way to a part one could lay up.

2. Governing models and the orientation lever

All three regimes share the plane-stress fiber constitutive law. With the engineering Voigt strain $\boldsymbol{\varepsilon}(\mathbf{u}) = [\,\partial_x u_x,\ \partial_y u_y,\ \partial_y u_x + \partial_x u_y\,]^{\mathsf{T}}$, the *rotated* orthotropic stiffness is

$$\mathbf{D}_f(\theta) = \mathbf{T}(\theta)\,\mathbf{Q}_f\,\mathbf{T}(\theta)^{\mathsf{T}}, \qquad \mathbf{T}(\theta) = \begin{bmatrix} c^2 & s^2 & -2sc \\ s^2 & c^2 & 2sc \\ sc & -sc & c^2 - s^2 \end{bmatrix}, \quad (c,s) = (\cos\theta, \sin\theta), \tag{1}$$

with $\mathbf{Q}_f$ the orthotropic fiber stiffness in material axes (nonzero entries $Q_{11} = E_{f1}/(1-\nu_{12}\nu_{21})$, $Q_{22} = E_{f2}/(1-\nu_{12}\nu_{21})$, $Q_{12} = \nu_{12}Q_{22}$, $Q_{66} = G_{12}$). We use the reference carbon-fiber material $E_{f1} = 131$, $E_{f2} = 9$, $G_{12} = 5$, $\nu_{12} = 0.27$ (contrast ≈ 14.6), $\rho = 1.6$. The scalar signature of the rotated tensor (1) is the off-axis Young's modulus, which it reproduces exactly,

$$\frac{1}{E_x(\theta)} = \frac{\cos^4\theta}{E_{f1}} + \frac{\sin^4\theta}{E_{f2}} + \Big(\frac{1}{G_{12}} - \frac{2\nu_{12}}{E_{f1}}\Big)\sin^2\theta\cos^2\theta, \tag{2}$$

running from 131 along the fiber to 9 across (matched to machine precision, Table 2) — the anisotropy the optimizer steers. The orientation $\theta(\mathbf{x})$ is the sole design field; the topology is fixed by the device geometry.

Three physics, one tensor.. On a domain $\Omega \subset \mathbb{R}^2$ we consider: (i) time-harmonic in-plane motion $\mathbf{u}\,e^{-\mathrm{i}\omega t}$,

$$-\nabla\cdot\big[(1+\mathrm{i}\eta)\,\mathbf{D}_f(\theta)\boldsymbol{\varepsilon}(\mathbf{u})\big] - \omega^2\rho\,\mathbf{u} - \mathrm{i}\,\omega\,c_{\mathrm{s}}\,\mathbf{u} = \mathbf{f}, \tag{3}$$

with small structural damping $\eta \in [0.02, 0.03]$ and an absorbing sponge $c_{\mathrm{s}} \geq 0$; (ii) static equilibrium $-\nabla\cdot(\mathbf{D}_f(\theta)\boldsymbol{\varepsilon}(\mathbf{u})) = \mathbf{f}$ under a service load; and (iii) the clamped generalized eigenproblem $\mathbf{K}(\theta)\boldsymbol{\varphi} = \lambda\mathbf{M}\boldsymbol{\varphi}$, $\lambda = \omega_n^2$, for natural frequencies. All three carry the design only through $\mathbf{K}(\theta) = \int_\Omega \boldsymbol{\varepsilon}(\cdot)^{\mathsf{T}}\mathbf{D}_f(\theta)\boldsymbol{\varepsilon}(\cdot)\,\mathrm{d}\mathbf{x}$.

A trigonometric-polynomial structure.. Expanding (1) in θ gives the exact five-term decomposition

$$\mathbf{D}_f(\theta) = \mathbf{C}_0 + \cos 2\theta\,\mathbf{C}_1 + \sin 2\theta\,\mathbf{C}_2 + \cos 4\theta\,\mathbf{C}_3 + \sin 4\theta\,\mathbf{C}_4, \tag{4}$$

with the constant $\mathbf{C}_k$ fixed linear combinations of $Q_{11}, Q_{12}, Q_{22}, Q_{66}$. Two consequences recur. First, $\partial\mathbf{D}_f/\partial\theta = -2\sin 2\theta\,\mathbf{C}_1 + 2\cos 2\theta\,\mathbf{C}_2 - 4\sin 4\theta\,\mathbf{C}_3 + 4\cos 4\theta\,\mathbf{C}_4$ is closed form — no finite differencing of the material tensor. Second, the anisotropy lives entirely in $\mathbf{C}_1, \ldots, \mathbf{C}_4$:

Proposition 1 (Isotropy annihilates the orientation lever)**.** *If $\mathbf{Q}_f$ is isotropic ($Q_{11} = Q_{22}$, $Q_{66} = \frac{1}{2}(Q_{11} - Q_{12})$) then $\mathbf{C}_1 = \cdots = \mathbf{C}_4 = \mathbf{0}$; hence $\mathbf{D}_f(\theta) \equiv \mathbf{C}_0$ and $\partial\mathbf{D}_f/\partial\theta \equiv \mathbf{0}$ for every θ.*

Proof. The stiffness of an isotropic material is an isotropic tensor, invariant under $\mathbf{T}(\theta)$: $\mathbf{T}\mathbf{Q}_f\mathbf{T}^{\mathsf{T}} = \mathbf{Q}_f$ for all θ. As (4) holds for all θ and $\{1, \cos 2\theta, \sin 2\theta, \cos 4\theta, \sin 4\theta\}$ are linearly independent, matching coefficients forces $\mathbf{C}_1 = \cdots = \mathbf{C}_4 = \mathbf{0}$. □

Because *every* design gradient below is built from $\partial \mathbf{D}_f/\partial\theta$ (§3), it must vanish identically when the material is set isotropic — a one-line exact regression test for any orientation-sensitivity code, which we use throughout.

Conforming voids.. Wave and structural devices are defined around holes. The density (SIMP) expedient fills a hole with an ersatz stiffness $E_{\min}$ so the "void" still supports a field. We instead mesh every hole *conformingly*: its boundary is a true traction-free edge and its interior is not part of Ω.

Proposition 2 (Void energy)**.** *For a hole V with traction-free boundary excluded by conforming meshing, the elastic and kinetic energy stored in V is exactly zero. If instead V is retained with ersatz stiffness $E_{\min}$ and mass $\rho_{\min}$, the interior field is generically nonzero and the spurious energy is $O(E_{\min})$, polluting any objective or gradient integrated over a region overlapping V.*

Conforming exclusion makes $V \cap \Omega = \varnothing$, so no degrees of freedom live in V; for soft-void the interior strain energy scales linearly with $E_{\min}$. A soft-void cloak can thus report cloaking that is partly the ersatz interior cancelling itself. All results below use conforming voids.

3. One lever, three physics: sensitivities

Discretizing any of the three models gives a linear (or generalized-eigen) problem whose design dependence is confined to $\mathbf{K}(\theta)$. Each objective's gradient reduces to a single bilinear form in $\partial\mathbf{D}_f/\partial\theta$, so Proposition 1 applies to all three.

Harmonic (cloak, lens).. A lens maximizes the energy in a focal region, $E = \int_{\Omega_{\text{reg}}} |\mathbf{u}|^2 \,\mathrm{d}\mathbf{x}$; a cloak minimizes the far-field mismatch to a void-free reference, $J = \int_{\Omega_{\text{obs}}} |\mathbf{u} - \mathbf{u}_{\text{ref}}|^2 \,\mathrm{d}\mathbf{x}$. Both are quadratic in the discrete state U. Adjoin the forward problem $\mathcal{A}(\theta)\mathsf{U} - \mathbf{f} = \mathbf{0}$ with a multiplier $\mathbf{\Lambda}$, $\mathcal{L} = J(\mathsf{U}) + \mathbf{\Lambda}^{\mathsf{T}}(\mathcal{A}\mathsf{U} - \mathbf{f})$; on the constraint manifold $\mathcal{L} = J$, and the total derivative separates into implicit and explicit parts,

$$\frac{\mathrm{d}J}{\mathrm{d}\theta_\ell} = \underbrace{\left(\tfrac{\partial J}{\partial \mathsf{U}} + \mathcal{A}^{\mathsf{T}}\mathbf{\Lambda}\right)^{\mathsf{T}} \tfrac{\mathrm{d}\mathsf{U}}{\mathrm{d}\theta_\ell}}_{\text{eliminated by the adjoint}} + \mathbf{\Lambda}^{\mathsf{T}} \frac{\partial \mathcal{A}}{\partial \theta_\ell}\mathsf{U}. \tag{5}$$

Choosing $\mathbf{\Lambda}$ to annihilate the implicit-state term gives the discrete adjoint equation $\mathcal{A}(\theta)^{\mathsf{T}}\mathbf{\Lambda} = -\partial J/\partial \mathsf{U}$, after which the design derivative is the explicit θ-dependence of the residual alone. Writing the θ-derivative bilinear form $b(\mathbf{u}, \mathbf{w}) \equiv \boldsymbol{\varepsilon}(\mathbf{u})^{\mathsf{T}}(\partial\mathbf{D}_f/\partial\theta)\boldsymbol{\varepsilon}(\mathbf{w})$ and using that only stiffness carries θ,

$$\frac{\mathrm{d}J}{\mathrm{d}\theta_\ell} = \int_{\Omega_\ell} \Big[b(\mathbf{u}_r, \boldsymbol{\lambda}_r) + b(\mathbf{u}_i, \boldsymbol{\lambda}_i) + \eta\big(b(\mathbf{u}_r, \boldsymbol{\lambda}_i) - b(\mathbf{u}_i, \boldsymbol{\lambda}_r)\big) \Big] \,\mathrm{d}\mathbf{x}, \tag{6}$$

with $(\mathbf{u}_r, \mathbf{u}_i), (\boldsymbol{\lambda}_r, \boldsymbol{\lambda}_i)$ the real/imaginary state and adjoint. The adjoint is driven by the objective's state-gradient $\partial J/\partial \bar{\mathsf{U}}$: with the region Gram matrix $\mathbf{M}_R = \int_{\Omega_R} \mathbf{N}^{\mathsf{T}}\mathbf{N}$, the focal-energy objective $E = \mathsf{U}^{\mathsf{H}}\mathbf{M}_{\text{reg}}\mathsf{U}$ that the lens *maximizes* ($J = -E$) has load $-\mathbf{M}_{\text{reg}}\mathsf{U}$ — simply the state restricted to the focal region, so the lens is self-adjoint up to that restriction — while the cloak mismatch $J = (\mathsf{U} - \mathsf{U}_{\text{ref}})^{\mathsf{H}}\mathbf{M}_{\text{obs}}(\mathsf{U} - \mathsf{U}_{\text{ref}})$ has load $\mathbf{M}_{\text{obs}}(\mathsf{U} - \mathsf{U}_{\text{ref}})$. The adjoint solve reuses the transposed forward factorization, so it adds no assembly.

Compliance (stiffness).. Minimizing the service compliance $c = \mathbf{f}^{\mathsf{T}}\mathbf{u}$ is self-adjoint, so

$$\mathrm{d}c/\mathrm{d}\theta_\ell = -\int_{\Omega_\ell} \boldsymbol{\varepsilon}(\mathbf{u})^{\mathsf{T}}(\partial \mathbf{D}_f/\partial\theta)\boldsymbol{\varepsilon}(\mathbf{u})\,\mathrm{d}\mathbf{x}.$$

Eigenfrequency (dynamics).. For a simple eigenvalue $\lambda_n = \omega_n^2$ with a mass-normalized mode $\boldsymbol{\varphi}_n$ ($\boldsymbol{\varphi}_n^{\mathsf{T}}\mathbf{M}\boldsymbol{\varphi}_n = 1$), first-order (Rayleigh -quotient) perturbation gives the design derivative

$$\frac{\partial\lambda_n}{\partial p} = \boldsymbol{\varphi}_n^{\mathsf{T}}\Big(\frac{\partial\mathbf{K}}{\partial p} - \lambda_n\frac{\partial\mathbf{M}}{\partial p}\Big)\boldsymbol{\varphi}_n, \qquad p \in \{\theta_\ell,\ z_e\}, \tag{7}$$

requiring no eigenvector derivative — the eigenvector's projection onto its own mode cannot change the quotient to first order. This is the identical generalized-eigenproblem sensitivity used throughout the companion study; the two settings here are just specializations. For an *orientation* variable only the stiffness carries the design ($\partial\mathbf{M}/\partial\theta = 0$), so (7) collapses to the strain-energy form $\mathrm{d}\lambda_n/\mathrm{d}\theta_\ell = \int_{\Omega_\ell}\boldsymbol{\varepsilon}(\boldsymbol{\varphi}_n)^{\mathsf{T}}(\partial\mathbf{D}_f/\partial\theta)\boldsymbol{\varepsilon}(\boldsymbol{\varphi}_n)\,\mathrm{d}\mathbf{x}$; a *density* variable additionally contributes the mass term $-\lambda_n(\rho_f - \rho_m)\,\boldsymbol{\varphi}_n^{\mathsf{T}}\mathbf{m}_e\boldsymbol{\varphi}_n$. The reading is clean — change in the mode's *strain* energy minus λ_n times the change in its *kinetic* energy — so a natural frequency rises when a perturbation stiffens the mode faster than it mass-loads it, and falls otherwise; everything follows from the one eigenpair already computed. The periodic (Bloch) band-structure sensitivity in §6.7 is the same expression with $\boldsymbol{\varphi}_n$ replaced by the reduced Bloch mode $\boldsymbol{\psi}_n = \mathbf{T}(\mathbf{k})\boldsymbol{\varphi}_n$ and $^{\mathsf{T}} \to {}^{\mathsf{H}}$.

Peak stress (strength).. The bracket's strength is governed by a p-norm surrogate of the peak von Mises stress,

$$\Sigma_p = \Big(\sum_e \tfrac{A_e}{\sum_k A_k}\,\tilde{\sigma}_e^{\,p}\Big)^{1/p}, \qquad \tilde{\sigma}_e = z_e^{\,q}\sqrt{\boldsymbol{\sigma}_e^{\mathsf{T}}\mathbf{V}\boldsymbol{\sigma}_e}, \quad \boldsymbol{\sigma}_e = \mathbf{D}_f(\theta_e)\,\mathbf{B}_e\mathbf{u}_e, \tag{8}$$

with $\mathbf{V}$ the plane-stress von Mises matrix and $z_e^{\,q}$ ($q = \frac{1}{2}$) a qp-relaxation that stops void elements carrying spurious stress. Its derivative has an explicit part (the element stress depends on θ through $\mathbf{D}_f$) and an implicit part through $\mathbf{u}$; the latter is a single adjoint solve $\mathbf{K}\boldsymbol{\mu} = \partial\Sigma_p/\partial\mathbf{u}$ reusing the static factorization, so $\partial\Sigma_p/\partial\theta_\ell = \partial\Sigma_p/\partial\theta_\ell|_{\text{expl}} - \boldsymbol{\mu}^{\mathsf{T}}(\partial\mathbf{K}/\partial\theta_\ell)\mathbf{u}$, again a $\partial\mathbf{D}_f/\partial\theta$ form. Both stress derivatives pass the finite-difference check to $\lesssim 10^{-7}$.

Each of (6) and its static/modal counterparts is built *only* from $\partial\mathbf{D}_f/\partial\theta$; setting the material isotropic must return an identically zero gradient (Proposition 1). We verify the non-trivial gradients against central finite differences to $\sim 10^{-6}$ and the isotropy identity to machine precision (Table 2).

4. Manufacturability, and the front

Smooth orientation field.. A raw per-cell orientation is mesh-scale noisy — unmanufacturable and, driving a wave field, incoherent. We parametrize θ by a coarse grid of support-point angles $\hat{\theta} = \{\hat{\theta}_i\}$ at sites $\{\mathbf{s}_i\}$ (spacing h_s), interpolated to the cell centroids $\{\mathbf{x}_c\}$ by a compactly-supported Wendland C^2 radial basis $\phi(q) = (1-q)^4(4q+1)$ (support $q < 1$):

$$\theta_c = (\mathbf{B}\hat{\theta})_c = \sum_i B_{ci}\hat{\theta}_i, \qquad B_{ci} = \frac{\phi(q_{ci})}{\sum_j \phi(q_{cj})}, \qquad q_{ci} = \frac{\|\mathbf{x}_c - \mathbf{s}_i\|}{r}, \tag{9}$$

a row-stochastic map ($\sum_i B_{ci} = 1$). By the chain rule the design gradient is a single transpose apply, $\partial J/\partial\hat{\theta} = \mathbf{B}^{\mathsf{T}}(\partial J/\partial\theta)$. The support radius r — not the mesh h — sets the finest resolvable fiber curvature, so the constraint below is mesh-independent, and the design is low-dimensional and smooth *by construction*. Where the device is mirror-symmetric (the lens about $y = L_y/2$) we enforce it exactly: a director is symmetric iff $\theta(x, L_y - y) = -\theta(x, y)$, so we use a symmetric support grid and project $\hat{\theta} \mapsto \frac{1}{2}(\hat{\theta} - \hat{\theta}_{\mathrm{mirror}})$ each iteration, removing the spurious asymmetry the triangulation would otherwise introduce.

Fiber-path curvature.. A printable tow has minimum turning radius R. The relevant quantity is the curvature of the fiber *streamline*: the directional derivative of θ along the fiber $\mathbf{n} = (\cos\theta, \sin\theta)$,

$$\zeta(\mathbf{x}) = (\mathbf{n}\cdot\nabla)\theta = \cos\theta\,\partial_x\theta + \sin\theta\,\partial_y\theta, \qquad |\zeta(\mathbf{x})| \le \zeta_{\mathrm{all}} = \tfrac{1}{R}. \tag{10}$$

The spatial gradients come *analytically* from the map (9): writing $W_{ci} = \phi(q_{ci})$, $S_c = \sum_j W_{cj}$ and differentiating the kernel *and* its normalization consistently,

$$(\mathbf{B}_x)_{ci} = \frac{\partial_x W_{ci}}{S_c} - B_{ci}\frac{\sum_j \partial_x W_{cj}}{S_c}, \qquad \partial_x W_{ci} = -\frac{20\,q_{ci}(1-q_{ci})^3}{r}\,\frac{x_c - s_{ix}}{\|\mathbf{x}_c - \mathbf{s}_i\|}, \tag{11}$$

and likewise $\mathbf{B}_y$ (using $\phi'(q) = -20\,q(1-q)^3$), so $\partial_x\theta = \mathbf{B}_x\hat{\theta}$, $\partial_y\theta = \mathbf{B}_y\hat{\theta}$, and both ζ and $\partial\zeta/\partial\hat{\theta}$ are exact — no finite differencing of the curvature.

From orientation to laid tows.. The orientation field is delineated into finite-pitch *toolpaths* by the wave-projection scheme: from the sign-free double-angle director $(\cos 2\theta, \sin 2\theta)$ a seam-free branch of θ is recovered by region-growing sign propagation, and a phase field ψ is solved for whose gradient is unit and orthogonal to the fibers,

$$\psi = \arg\min_{\psi} \int_{\Omega} \left\| \nabla\psi - [-\sin\theta,\ \cos\theta]^{\mathsf{T}} \right\|^2 \mathrm{d}\mathbf{x}, \tag{12}$$

matched *directly* (a gradient least-squares, not the normal-equations Poisson form, which would collapse the pitch). The evenly-spaced level sets of ψ — the peaks of $\chi = \frac{1}{2} + \frac{1}{2}\cos(2\pi\psi/d)$ — are the as-manufactured tows at pitch d; they follow the fiber direction, keep a fixed spacing, and are masked out of every void, so no tow crosses a hole. The tow figures in §6 are these level sets.

The front.. We impose (10) with a smooth one-sided penalty $\Pi = \langle \max(0, \zeta^2 - \zeta_{\mathrm{all}}^2)\rangle$ and compose the total gradient by pulling the objective sensitivity back through the CS-RBF transpose and balancing it against the curvature penalty,

$$\mathbf{g} = \mathbf{B}^{\mathsf{T}}\frac{\partial\mathcal{J}}{\partial\theta} + \mu\,\frac{\|\mathbf{B}^{\mathsf{T}}\partial\mathcal{J}/\partial\theta\|}{\|\partial\Pi/\partial\hat{\theta}\|}\,\frac{\partial\Pi}{\partial\hat{\theta}}, \tag{13}$$

with the penalty pre-scaled to the objective-gradient norm. The weight μ is adapted toward the limit — raised while $\max|\zeta| > \zeta_{\mathrm{all}}$, eased otherwise — and the best iterate that is actually feasible ($\max|\zeta| \le \zeta_{\mathrm{all}}$) is retained, so the reported design is guaranteed printable (Algorithm 1, Figure 1). Sweeping ζ_{all} is an ε-constraint traversal of the performance–curvature Pareto: the *manufacturability front*, whose far ($\zeta_{\mathrm{all}} \to \infty$) end is the unconstrained optimum (or, in the cloak, the singular transformation ideal) and whose near end is the gently-steered, easily-printed design. The curve — not any single optimum — is the design object.

Optimizer.. Each step, the Method of Moving Asymptotes replaces $\mathcal{J}$ and Π by convex separable approximations built from moving asymptotes $L_\ell < \hat{\theta}_\ell < U_\ell$ that contract or expand by the sign history of $\hat{\theta}_\ell$, with a per-step move limit; the box $|\hat{\theta}_\ell| \le \frac{\pi}{2}$ reflects the director periodicity. Because raw wave objectives are $O(10^{-4})$, we feed MMA the scale-invariant log-gradient $\partial_\theta \ln E$ (lens) or $\partial_\theta \ln J$ (cloak).

Algorithm 1 Curvature-constrained fiber-orientation design (best-feasible; cf. Figure 1)

Require: mesh with conforming voids, material $\mathbf{Q}_f$, objective (harmonic / static / eigen), curvature limit ζ_{all}, support sites $\{\mathbf{s}_i\}$ and radius r

build $\mathbf{B}, \mathbf{B}_x, \mathbf{B}_y$ (9)–(11) and the symmetry projector $\mathbf{S}$; $\hat{\theta} \leftarrow \mathbf{0}$ ▷ straight-fiber baseline
(cloak only) solve the void-free reference $\mathbf{u}_{\text{ref}}$ on the same mesh
for $\mu \in \{\mu_0, \dots, \mu_{\max}\}$ **do** ▷ adaptive penalty continuation
  for $t = 1, \dots, T$ **do**
    $\hat{\theta} \leftarrow \mathbf{S}\hat{\theta}$; $\theta \leftarrow \mathbf{B}\hat{\theta}$ ▷ symmetrize, then interpolate
    assemble $\mathcal{A}(\theta)$ (real 4-field (14)) *or* $\mathbf{K}, \mathbf{M}$; factor, and **solve** the state
    evaluate $\mathcal{J}$; **adjoint** $\mathcal{A}^{\mathsf{T}}\mathbf{\Lambda} = -\partial\mathcal{J}/\partial\mathsf{U}$ ▷ reuse factors
    $\partial\mathcal{J}/\partial\theta \leftarrow$ (6) / (7) / (8) ▷ a $\partial\mathbf{D}_f/\partial\theta$ form
    $\zeta \leftarrow \cos\theta\, \mathbf{B}_x\hat{\theta} + \sin\theta\, \mathbf{B}_y\hat{\theta}$; $\mathbf{g} \leftarrow$ (13); $\hat{\theta} \leftarrow \mathrm{MMA}(\hat{\theta}, \mathbf{g};\ |\hat{\theta}| \le \frac{\pi}{2})$
  end for
  retain the best feasible iterate; **if** $\max_c |\zeta_c| \le \zeta_{\text{all}}$ **break** ▷ printable
end for
delineate toolpaths (12); **return** best design, fields, and tows

5. Numerical formulation and verification

Real four-field split.. The weak form of (3) is complex-symmetric, $[(1+\mathrm{i}\eta)\mathbf{K} - \omega^2\mathbf{M} + \mathrm{i}\omega\mathbf{C}]\mathbf{u} = \mathbf{f}$. Writing the complex operator as $A + \mathrm{i}B$ with $A = \mathbf{K} - \omega^2\mathbf{M}$ and $B = \eta\mathbf{K} + \omega\mathbf{C}$ both real symmetric, and $\mathbf{u} = \mathbf{u}_r + \mathrm{i}\mathbf{u}_i$, the equivalent real block system on $(u_{xr}, u_{yr}, u_{xi}, u_{yi})$ is

$$\begin{bmatrix} A & -B \\ B & A \end{bmatrix} \begin{bmatrix} \mathbf{u}_r \\ \mathbf{u}_i \end{bmatrix} = \begin{bmatrix} \mathbf{f}_r \\ \mathbf{f}_i \end{bmatrix}, \tag{14}$$

solved by a real `PETSc` build. Block (14) is exactly $[\mathrm{Re}, -\mathrm{Im}; \mathrm{Im}, \mathrm{Re}]$ of the complex operator: expanding $(A + \mathrm{i}B)(\mathbf{u}_r + \mathrm{i}\mathbf{u}_i) = (A\mathbf{u}_r - B\mathbf{u}_i) + \mathrm{i}(B\mathbf{u}_r + A\mathbf{u}_i) \stackrel{!}{=} \mathbf{f}_r + \mathrm{i}\mathbf{f}_i$ and equating real and imaginary parts separately gives the two coupled real systems, so no complex arithmetic is needed and the structural damping $\eta\mathbf{K}$ joins the sponge in the off-diagonal block. The elements are conforming constant-strain triangles; every hole is meshed as a true boundary (Proposition 2) with `gmsh`, not a soft inclusion on a structured grid — staircased boundaries would scatter the wave and corrupt the gradient.

Truncation, drive, and reference.. A finite plate is made to radiate as if unbounded by a mass-proportional sponge $c_{\mathrm{s}}(\mathbf{x})$ that ramps quadratically into a margin of width m on the three non-driven edges,

$$c_{\mathrm{s}}(\mathbf{x}) = c_0 \left[\max\left(0,\ \tfrac{x-(L_x-m)}{m},\ \tfrac{m-y}{m},\ \tfrac{y-(L_y-m)}{m}\right)\right]^2, \tag{15}$$

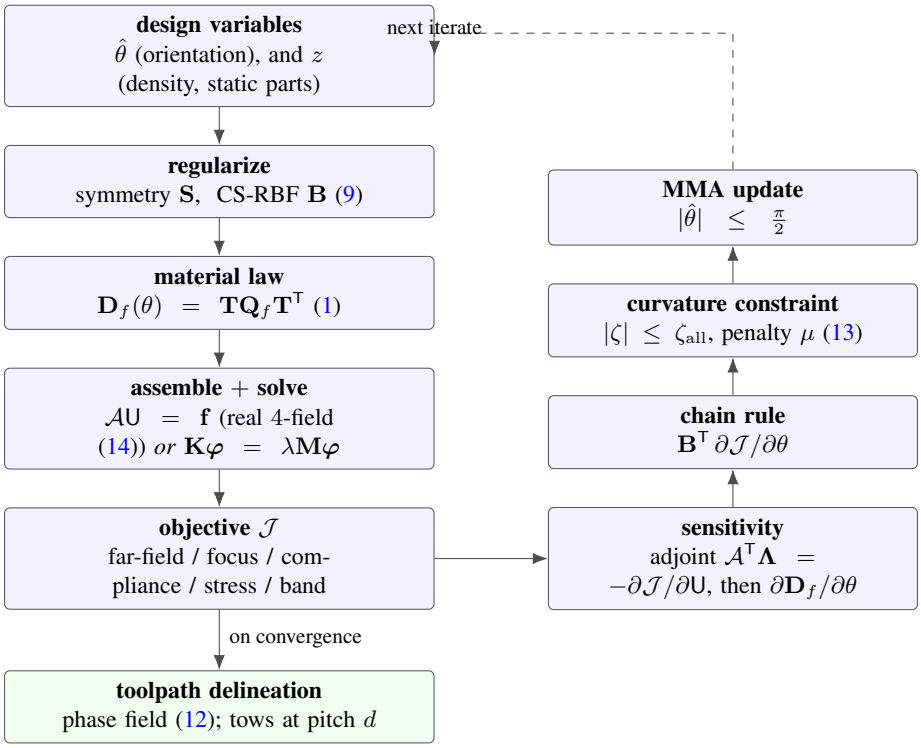


Figure 1: **Data flow of the design loop.** Left column: design variables descend through the regularization operators into the material law, the assembled operator, and the solve. Right column: the gradient ascends through the adjoint (wave, compliance, stress) or eigenvalue perturbation (modal, band), then the *transpose* of the CS-RBF map and the curvature constraint, to the MMA update, which returns the next iterate (dashed). Green: the toolpath delineation that turns the converged orientation field into printable tows.

driven by an x-directed line source $\mathbf{f} = (A\, e^{-\beta(x-x_0)^2}, 0)$ near the left edge (a plane pressure wave). Because $\mathbf{M}$, the sponge $\mathbf{C}$, and $\mathbf{f}$ carry no θ, they drop out of every sensitivity (6). The cloak reference $\mathbf{u}_{\mathrm{ref}}$ is the field solved on the *same* mesh with the void filled — the wave that would exist with no hole — so J measures exactly the scattering the void introduces. Table 1 lists the constants.

Table 1: Model and design parameters (nondimensional units).

Group	Parameter	Cloak / lens	Bracket
Material	$E_{f1}, E_{f2}, G_{12}, \nu_{12}$	$131,\ 9,\ 5,\ 0.27 \quad (\rho = 1.6)$	
Damping	structural η	0.02–0.03	
Wave	frequency ω; source (A, β, x_0)	50; $(30, 80, 0.1)$	—
Sponge	(c_0, m)	$(80, 0.6)/(70, 0.55)$	—
Domain	(L_x, L_y)	$(8, 5)$	$(3.5, 2.0)$
Cloak void	R_V, design shell R_{des}	$0.55,\ 1.5$	—
Orientation	CS-RBF spacing h_s, radius r	$(0.33, 0.7)/(0.28, 0.6)$	0.045–0.06
Curvature	limit $\zeta_{\mathrm{all}} = 1/R$	swept	9 (featured)

Verification.. Table 2 collects the checks that make the formulation trustworthy across regimes: the closed-form gradient against central finite differences, the isotropy-annihilation identity (Proposition 1), the rotated tensor against the analytic off-axis modulus, and mesh-refinement stability of the reported metrics.

Table 2: Verification of the orientation sensitivities and the model, across regimes.

Check	Quantity	Result
Finite-difference gradient	$\lvert\, \mathrm{d}J/\mathrm{d}\theta - \mathrm{FD} \,\rvert / \lvert \mathrm{FD} \rvert$	$\lesssim 10^{-6}$ (harmonic, static, modal)
Isotropy annihilation	$\max_\ell \lvert\, \mathrm{d}J/\mathrm{d}\theta_\ell \,\rvert$ with $\mathbf{Q}_f$ isotropic	$\sim 10^{-16}$ (machine zero)
Off-axis modulus	$E_x(\theta)$ from $\mathbf{D}_f(\theta)$ vs. analytic	machine precision
Conforming void	interior field energy (Prop. 2)	0 (vs. $O(E_{\min})$ soft-void)
Mesh refinement	reported gain/reduction over h	stable to plotting accuracy

Optimization and gradient checks.. The design vector is the CS-RBF support amplitudes $\hat{\boldsymbol{\theta}}$ (a few hundred per case, Table 1), reduced to a smooth per-cell field by the map $\theta = \Phi\hat{\boldsymbol{\theta}}$ and, for the bracket, co-designed with the SIMP density z; every case starts from straight fibers $\hat{\boldsymbol{\theta}} = \mathbf{0}$ (the honest baseline). The reduced gradient is assembled by the transpose chain of Figure 1 — adjoint or eigenvalue perturbation, then Φ^{T} and the curvature term (13) — and handed to the method of moving asymptotes (MMA) with a move limit of 0.08–0.10 on the box $|\hat{\theta}_\ell| \le \pi/2$ and adaptively updated asymptotes; because raw wave objectives are $O(10^{-4})$ we feed MMA the scale-invariant log-gradient. Mirror-symmetric cases (the two-focus splitter, the cloak) project both the iterate and the gradient onto the symmetric subspace so the design cannot drift off symmetry. Runs take 75–140 iterations under the adaptive curvature-penalty continuation and best-feasible safeguard of Algorithm 1; a stalled objective is perturbed by $\sigma = 0.1$ rad and resumed. Every non-trivial gradient is verified against central finite differences $[\, J(\hat{\boldsymbol{\theta}} + \varepsilon\mathbf{e}_i) - J(\hat{\boldsymbol{\theta}} - \varepsilon\mathbf{e}_i)\,]/2\varepsilon$ at $\varepsilon = 10^{-5}$ on randomly chosen support DoFs (relative error $\lesssim 10^{-6}$), and against the isotropy-annihilation identity, which forces *every* component to machine zero when $\mathbf{Q}_f$ is made isotropic (Table 2). This adjoint / eigenvalue-perturbation

/ MMA machinery follows the established stiffness–wave co-design methodology; the contribution here is the curvature-constrained, best-feasible *front*, not the sensitivity framework itself.

6. Results: one front, four regimes

The manufacturability front appears in every regime we tried — static stiffness and strength, wave scattering (cloaking), wave focusing, beam shaping, shielding, and routing, and periodic band structure — because all are driven by the same $\partial \mathbf{D}_f/\partial\theta$ lever. Table 3 distils the four headline instances: each has an unconstrained optimum (or, for stiffness and cloaking, an *ideal* that is unmanufacturable) and a printable design obtained by holding the tow curvature to a process limit, and in every case the printable design recovers a large, quantifiable share of the unconstrained performance. We take the regimes in turn.

Table 3: One manufacturability front, four regimes. Each printable design is the best-feasible optimum at the stated tow-curvature limit $\zeta_{\text{all}} = 1/R$.

Regime	Objective	Ideal / unconstrained	Printable	at ζ_{all}
Stiffness (bracket)	compliance gain vs. straight	64% pointwise / 60%	57%	9
Cloaking	far-field reduction J_0/J	singular / $36\times$	$9.6\times$	$1/R_V$
Focusing (lens)	focus gain E/E_0	— / $47\times$	$12\times$	2.8
Band structure	band flattening	— / —	$4.7\times$	—

6.1. The static front: stiffness and strength

A CFRP mounting bracket is clamped at two bolt holes and loaded on the rim of a third; we co-design its topology and fiber orientation to be stiff, strong, and printable. The anisotropy is a front (Figure 2): sweeping ζ_{all} at fixed topology and mass, the compliance gain over the best straight (0°) layup climbs from 37% (gentle tows) toward a *pointwise* ceiling of $\sim 64\%$ — the stiff axis aligned with the local principal stress everywhere, which is unmanufacturable because that direction turns sharply. The featured design is a deliberately aggressive but printable point ($|\zeta| \le 9$, mean steering $\sim 18^\circ$, gain 57%): a clean, fully load-bearing A-frame that is $9.4\times$ stiffer than an equal-mass isotropic plate at $0.14\times$ its peak stress, with its fundamental $2.98\times$ the plate's rising *proportionately* ($\omega_1^2 \approx$ stiffness). The same eigenvalue machinery also drives the first two natural frequencies together, closing ω_2/ω_1 from the plate's ~ 3.1 toward ~ 1.2 — a purely anisotropic effect.

Stiffness and the fundamental frequency are, however, not the same objective, and the bracket makes the distinction concrete. Weighting the modal term w_ω in the joint stiffness/eigenvalue design traces a stiffness–frequency Pareto front (Figure 4): the pure-stiffness A-frame ($w_\omega = 0$, the part we build) sits at $\sim 9.5\times$ stiffness and $\sim 3.2\times$ fundamental, essentially on the *proportionate* line $\omega_1^2 \propto$ stiffness. Turning w_ω up lifts ω_1 into a *disproportionate* region — peaking near $4.3\times$, well above what stiffness alone would buy — but only by morphing the clean truss into a thinner, modal-stiffened frame whose added material is a lightly-loaded resonance tuner rather than a load path, so the stiffness gain falls to 5–$7\times$. Because the front is smooth, any modal target between the ends is one weight away; we build the $w_\omega = 0$ A-frame, but the same curvature-feasible machinery reaches the whole front.

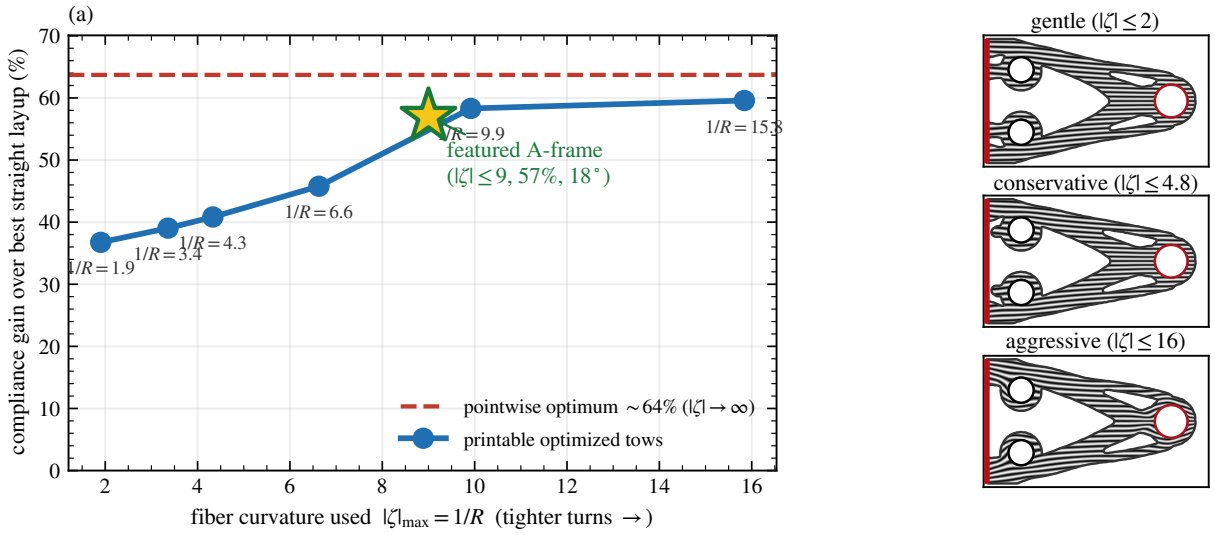


Figure 2: **The static front.** Compliance gain over the best straight layup versus the fiber curvature actually used, at fixed topology and mass. As the tow-curvature limit is relaxed the fibers steer harder (mean $9^\circ \to 20^\circ$) and the gain climbs from 37% toward the ∼64% pointwise optimum (dashed, unmanufacturable). The featured A-frame (star, $|\zeta| \le 9$) recovers 57% while staying printable; insets show the tow field at three points on the front.

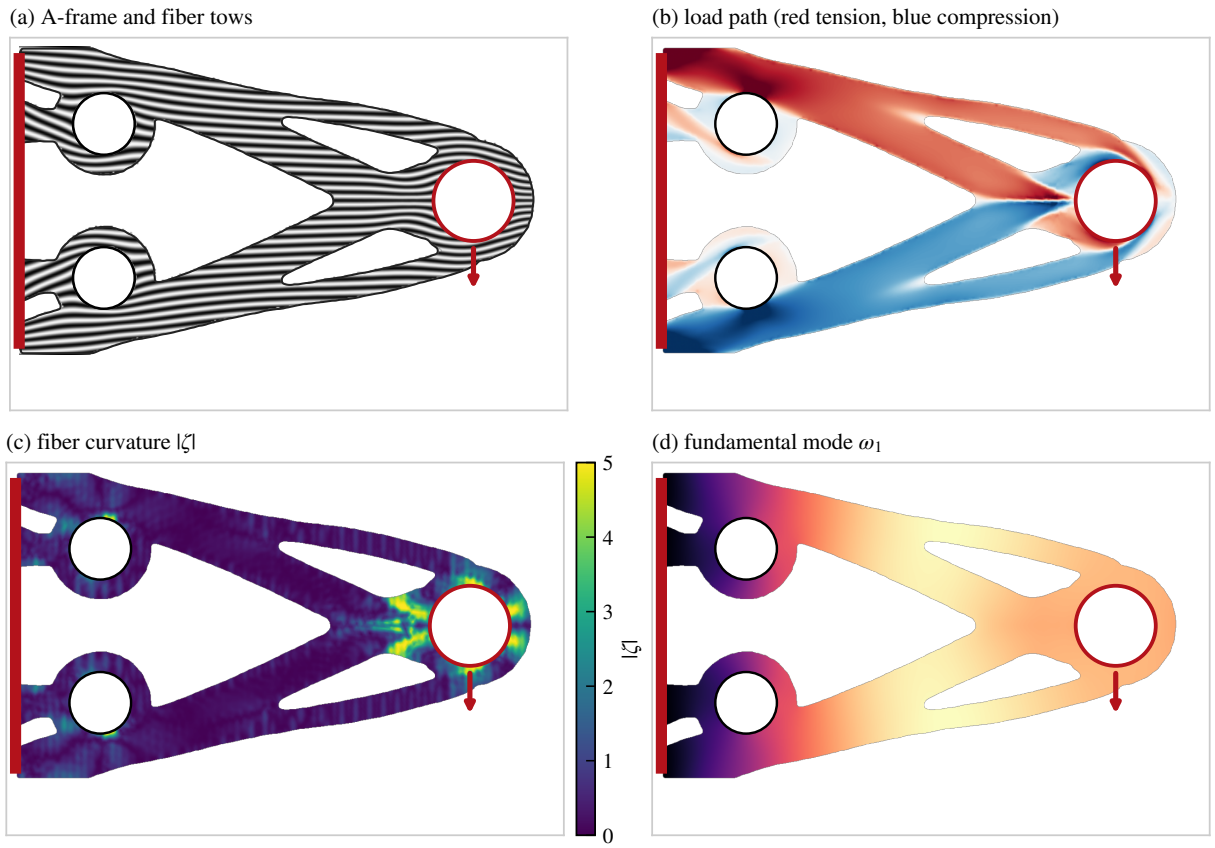


Figure 3: **The printable static design** (the star in Figure 2). (a) the clean A-frame, fibers wrapping every hole; (b) the load path — every member carries service load (red tension, blue compression), with no dead appendages; (c) fiber curvature $|\zeta|$, at the print limit only where the tows wrap the loaded ring; (d) the fundamental mode, $\omega_1 = 2.98\times$ the equal-mass plate.

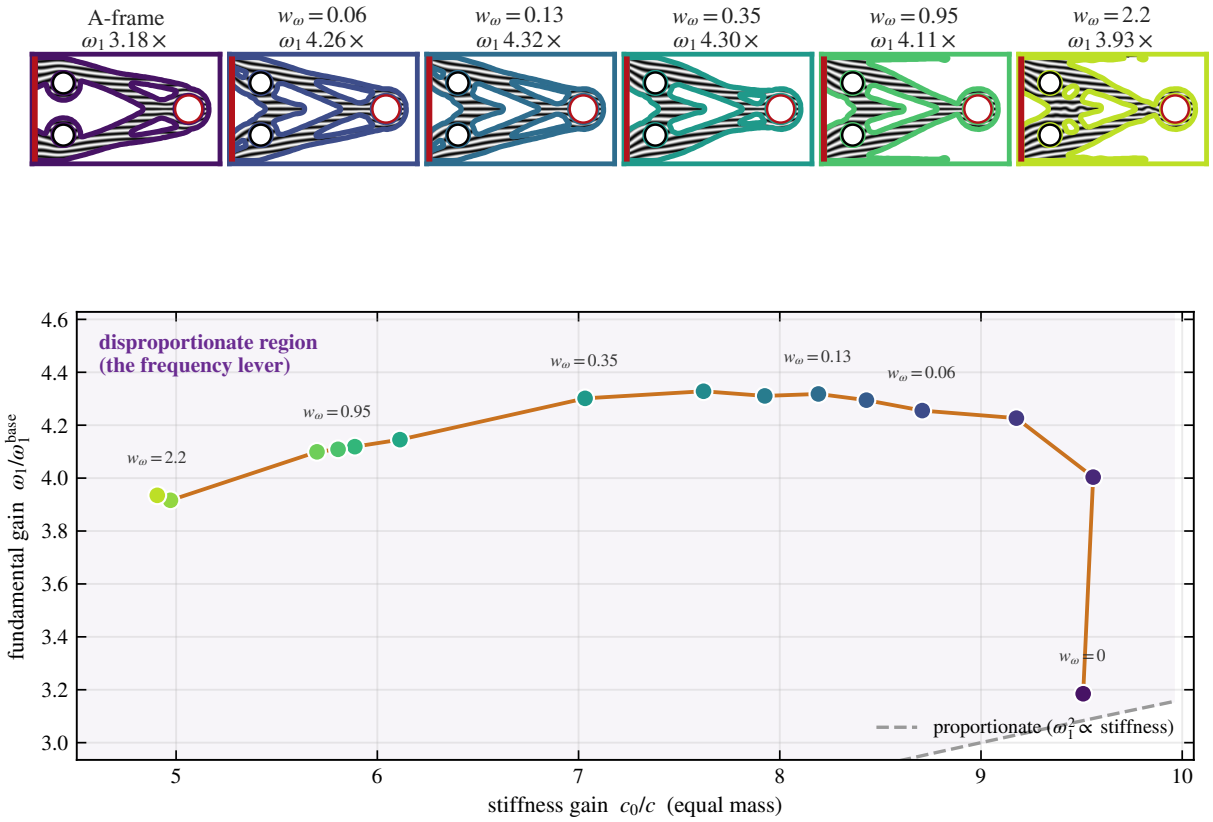


Figure 4: **The stiffness–frequency front (the frequency lever).** Raising the modal weight w_ω morphs the pure-stiffness A-frame ($w_\omega = 0$, the part we build) into a thinner, modal-stiffened truss (top row: topology and tows at representative weights). The fundamental gain $\omega_1/\omega_1^{\text{base}}$ climbs into the disproportionate region above the proportionate line $\omega_1^2 \propto$ stiffness (dashed), peaking near $4.3\times$, while the stiffness gain c_0/c falls from $\sim 9.5\times$ to $\sim 5\times$. A single joint co-design traces the whole front.

6.2. *The sharpest instance: cloaking a void*

A plane wave at $\omega = 50$ crosses a solid plate containing a conforming traction-free void of radius R_V; the design shell is the annulus $R_V < r < R_{\text{des}}$, with straight fibers outside. We minimize the far-field mismatch J to the void-free reference over an observation box behind the void. Uncloaked, the void casts a shadow of suppressed amplitude downstream and a standing pattern of reflections upstream (Fig. 5b); the cloak must restore the undisturbed plane wave (Fig. 5a). Cloaking is the sharpest point of the whole thesis because its ideal endpoint is not merely hard but *singular* (Figure 5).

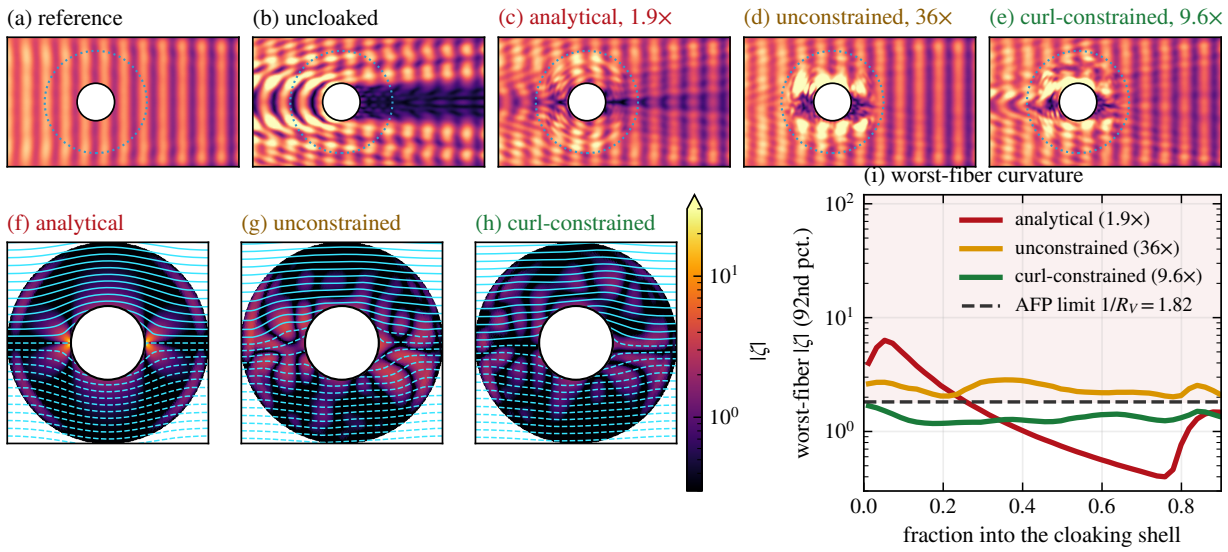


Figure 5: **The cloak in three acts** at $\omega = 50$. *Top* ($|u|$): (a) void-free reference; (b) uncloaked shadow and reflections; (c) the analytical transformation cloak, whose *directions* we impose on the bounded-contrast fiber — only $1.9\times$; (d) free optimization restores the far field $36\times$ but exceeds the print limit; (e) the curvature-constrained optimum keeps almost all of it ($9.6\times$), printable. *Bottom* (tow toolpaths over $|\zeta|$): (f) the analytical field is *singular* at the rim; (g) the free optimum is bounded but over the limit; (h) the constrained optimum lies under it; (i) worst-fiber curvature into the shell — only the constrained design stays beneath $1/R_V$ throughout.

The transformation recipe maps the void to a point and reads off the material an ideal cloak needs. For elastic waves that material is singular: the required anisotropy diverges at the inner surface [2]. A fiber, of fixed finite contrast, cannot supply it, and this surfaces twice. *Manufacturability:* mapping straight tows through the radial transformation forces $|\zeta| \to \infty$ at the rim (Fig. 5f) — the transformation cloak is literally the singular endpoint of the front. *Effectiveness:* even ignoring fabrication, imposing the transformation *directions* on the real fiber cloaks only $1.9\times$ (Fig. 5c), because the missing ingredient is the diverging *magnitude*, not the direction.

Free optimization — never attempting the singular ideal — finds a bounded-curvature field that restores the far field $36\times$ (Fig. 5d), roughly nineteen times the analytical cloak, but with tows above the print limit (Fig. 5g,i). Re-solving with the curvature constraint active (Algorithm 1) yields a genuinely printable cloak that still reduces the mismatch $9.6\times$ (Fig. 5e,h), about five times the analytical prescription. Sweeping ζ_{all} traces the cloak's front (Figure 6): reduction climbs smoothly with the allowed curvature from the gentle regime toward the unconstrained ceiling, with the analytical cloak marooned at the singular end where no

process reaches. The transformation cloak says cloaking is possible in principle; optimization on the front is what makes it a part one could lay up.

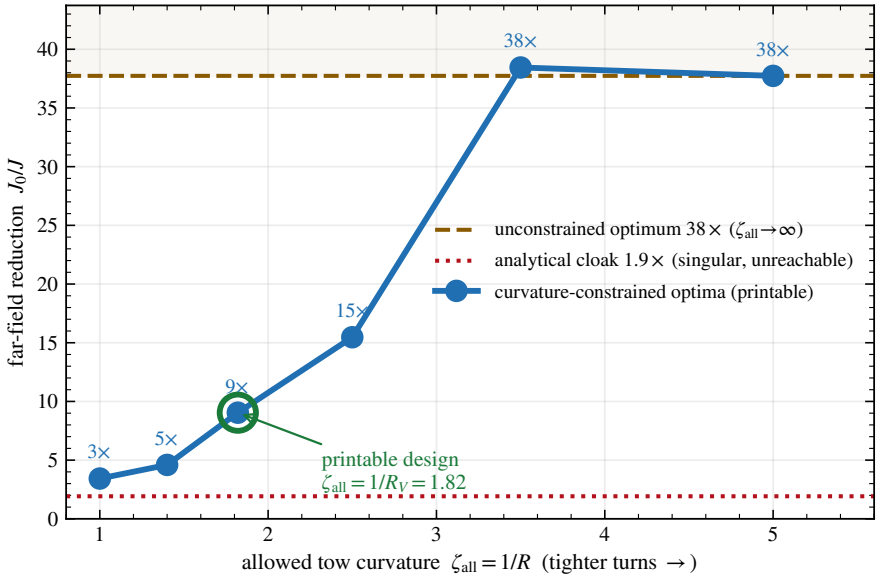


Figure 6: **The cloak's manufacturability front.** Far-field reduction J_0/J versus the allowed tow curvature $\zeta_{\mathrm{all}} = 1/R$, from an independent sweep (best-feasible optimum at each ζ_{all}). The constrained optima (curve) climb from the gentle regime and saturate at the unconstrained ceiling once the constraint is inactive; the geometry-tied limit $1/R_V$ marks the printable design used in Figure 5(e). The analytical transformation cloak ($1.9\times$) sits at the singular $\zeta_{\mathrm{all}} \to \infty$ end that no process can realize. Endpoints ($\sim 38\times$ ceiling, $\sim 9\times$ at $1/R_V$) agree with the longer-converged featured values ($36\times$, $9.6\times$) to within optimization variance.

6.3. The wave front: focusing

The same tension governs a curvilinear-fiber *lens* that steers an incident plane wave to a focus (objective $E = \int_{\Omega_{\mathrm{reg}}} |\mathbf{u}|^2 \, \mathrm{d}\mathbf{x}$). Free optimization concentrates the field with a focus gain of $47\times$ at peak curvature $|\zeta|_{\max} = 6.3$; enforcing $\zeta_{\mathrm{all}} = 2.8$ yields a printable lens with gentle, large-radius tows and a gain of $12\times$ (Figure 7). Cloak and lens thus sit on the same qualitative front — anisotropic wave control buys performance in near-proportion to the tow curvature a process allows.

6.4. Beam shaping: two foci and a null

The focus objective generalizes to several targets at once. We drive a localized aperture beam and optimize θ to place energy at two off-axis foci while holding a central null (Figure 8): straight fibers send the beam straight through the *intended null*, leaving the foci dark, whereas the optimized orientation forks it into two branches that light the foci and leave a dark centre — inverting the pattern into a foci/null contrast of $18\times$, a $\sim 1500\times$ swing from the straight-fiber baseline. The design maximizes the foci energy *fraction*: with region energies $E_R = \int_{\Omega_R} |\mathbf{u}|^2 \, \mathrm{d}\mathbf{x}$ and the whole-plate energy E_Ω, the objective is the normalized, null-penalized ratio

$$\mathcal{J}_{\mathrm{multi}} = \frac{E_A + E_B - \gamma \, E_{\mathrm{null}}}{E_\Omega}, \tag{16}$$

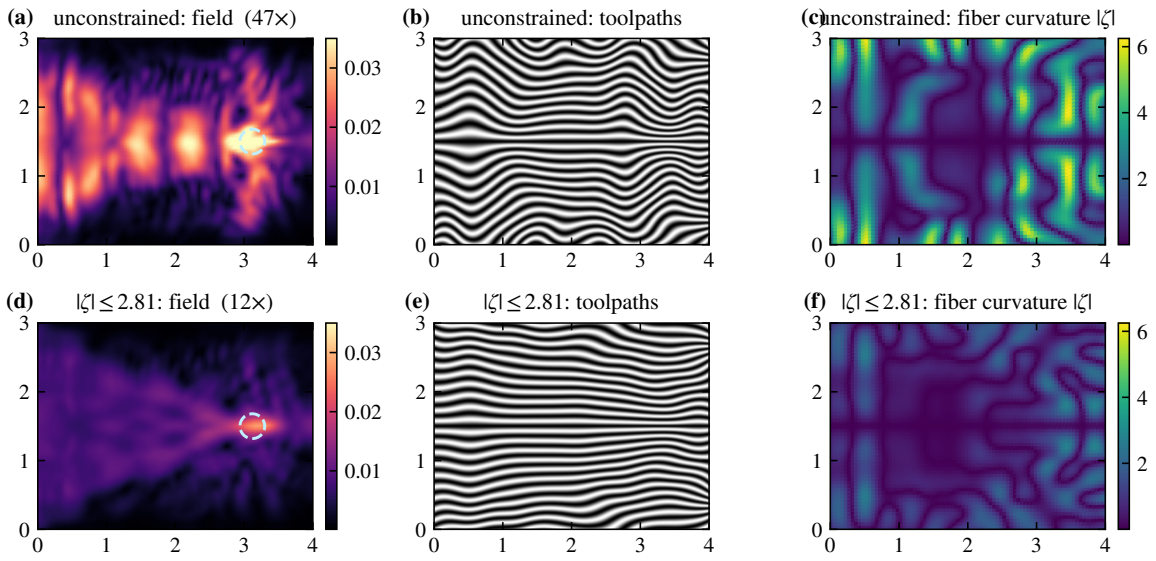


Figure 7: **The wave front for a lens.** *Top:* the unconstrained optimum focuses strongly (47×) with sharp tows ($|\zeta|_{\max} = 6.3$). *Bottom:* with the curvature constraint at $\zeta_{\text{all}} = 2.8$, the field still focuses (12×), the toolpaths are gentle printable arcs, and $|\zeta|$ is capped at the limit. The dashed circle in the field panels marks the focus target the lens is optimized to fill.

with γ a fixed null-suppression weight. Its gradient is the quotient rule

$$\frac{\mathrm{d}\mathcal{J}_{\text{multi}}}{\mathrm{d}\theta_\ell} = \frac{(\dot{E}_A + \dot{E}_B - \gamma\,\dot{E}_{\text{null}})\,E_\Omega - (E_A + E_B - \gamma\,E_{\text{null}})\,\dot{E}_\Omega}{E_\Omega^2}, \qquad \dot{E}_R \equiv \frac{\mathrm{d}E_R}{\mathrm{d}\theta_\ell}, \tag{17}$$

where each region rate $\dot{E}_R$ is the harmonic region-energy sensitivity (6) with its own adjoint load $\mathbf{M}_R\mathsf{U}$. Normalizing by E_Ω makes suppressing transmission useless — it shrinks numerator and denominator alike — so the optimizer cannot cheat and must genuinely relocate the incident beam into the foci. The device is a single solid plate; the beam splitting is entirely in the fiber pattern, and it sits on the same manufacturability front — a curvature limit trades focus contrast for gentler tows exactly as in §6.3.

6.5. Shielding: excluding energy in solid material

A complementary objective *excludes* energy from a region. On a solid plate with drilled through-holes, orienting the fibers to divert an incident wave around a protected zone makes it 3–5× quieter across a frequency band, while the *same* layup raises the static stiffness 24% and lowers the mass 15% (Figure 9): shielding and stiffness are co-designed, not traded. Because every hole is conforming (Proposition 2), the shielded region — and each void — carries provably zero interior energy, so the reported suppression is physical, not a soft-void artefact.

6.6. Routing: a curved delivery corridor

Routing steers energy *to* a destination while starving a bystanding region. Designing θ to deliver an incident wave to an exit port around an obstacle raises the exit energy 698× and confines it 97× downstream, while holding a guarded annulus at 0.17× its baseline energy — a protection factor of 4002× relative to the

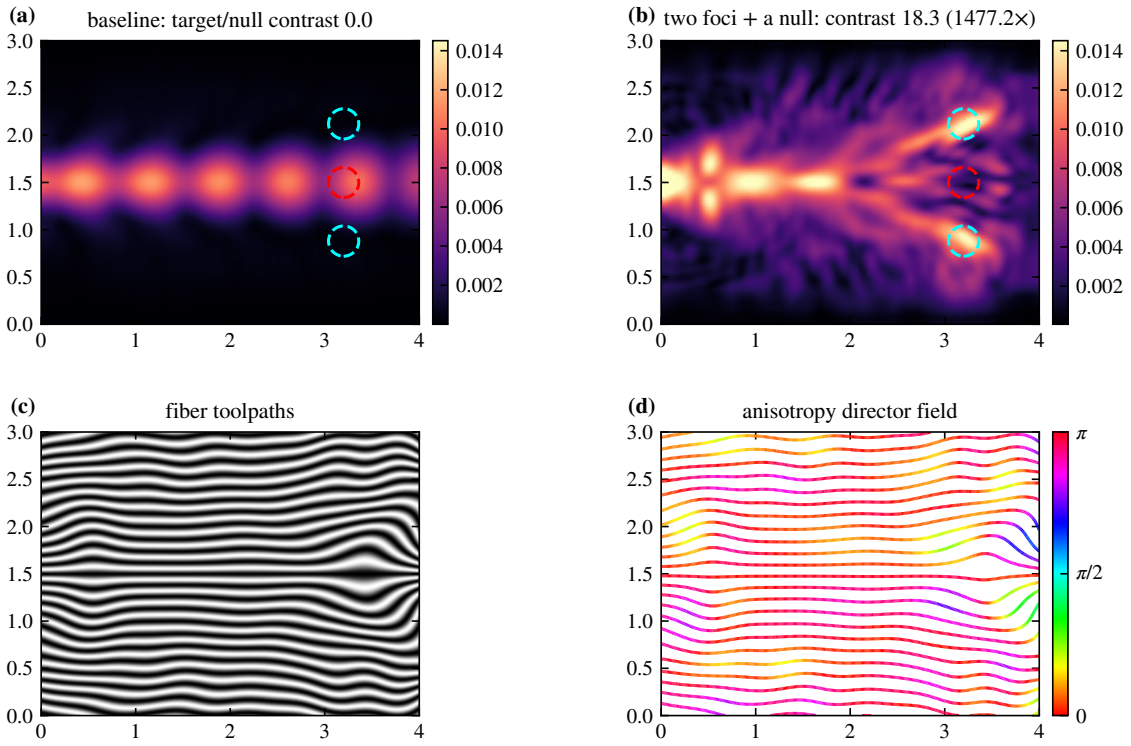


Figure 8: **Multi-target beam shaping.** Straight fibers (left) send the incident beam straight through the intended null, leaving the two off-axis foci (cyan) dark; the optimized orientation (right) forks the beam into the two foci with a dark centre (red null) — a foci/null contrast of $18\times$, all in a single solid sheet. Panels: baseline and optimized $|u|$, the fiber toolpaths, and the anisotropy director field.

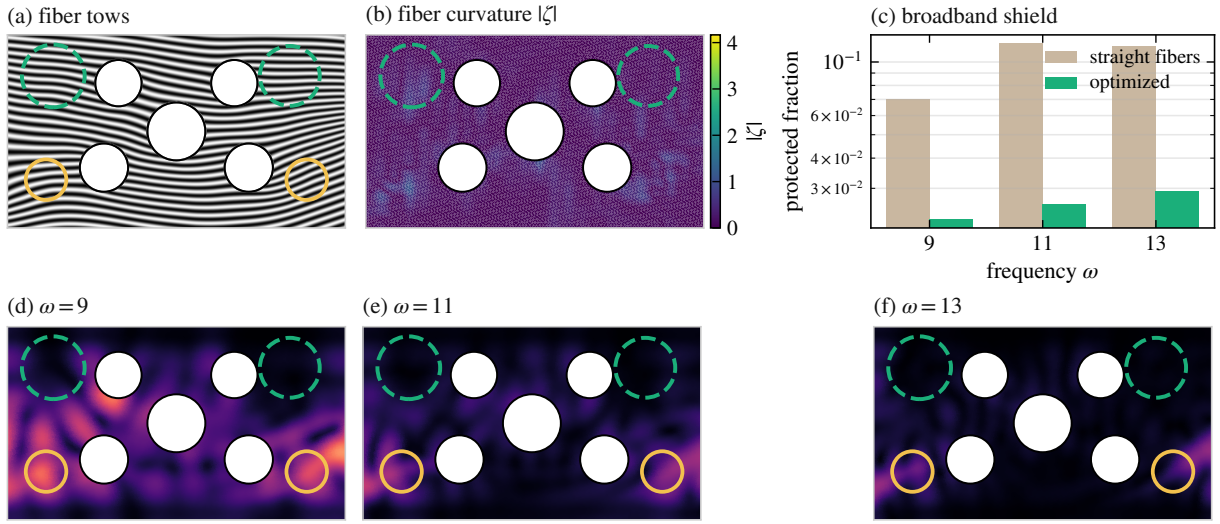


Figure 9: **A vibration shield in solid material.** Fiber orientation on a through-holed CFRP plate diverts incident energy around a protected region (3–$5\times$ quieter across the band) while the same layup is 24% stiffer at 15% lower mass. The holes are conforming, so the shielded zone holds zero wave energy.

straight-fiber layup (Figure 10). The corridor is a curved fiber channel in otherwise solid material, and its tow curvature is again capped at the print limit.

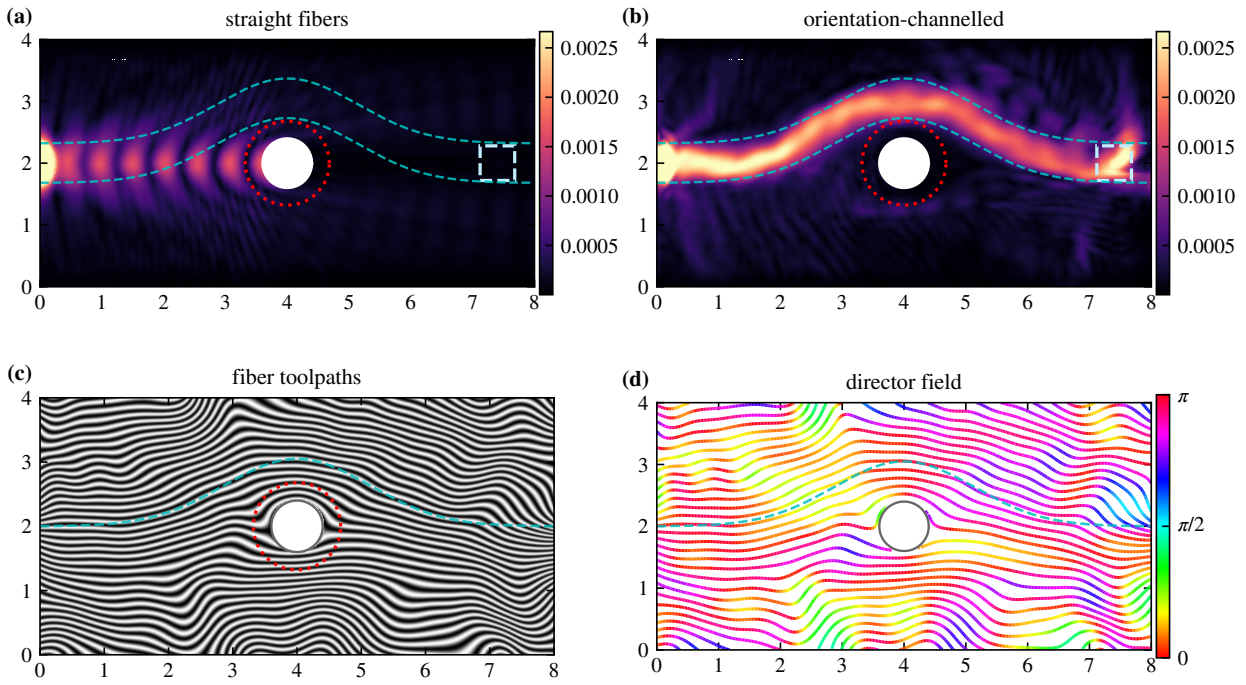


Figure 10: **Energy routing and joint protection.** The optimized fiber corridor delivers the incident wave to an exit port (698× exit-energy gain, 97× downstream confinement) while starving a guarded annulus (0.17× baseline energy), a protection factor of 4002×.

6.7. The periodic front: band structure

Finally, the same fiber tensor and curvature constraint act on a *Bloch cell*. On a periodic cell the state is a Bloch mode $\boldsymbol{\varphi}_n(\mathbf{k})$ of $\mathbf{K}(\theta)\boldsymbol{\varphi} = \lambda\mathbf{M}\boldsymbol{\varphi}$; the reduced matrices $\mathbf{K}_r = \mathbf{T}(\mathbf{k})^{\mathsf{H}}\mathbf{K}\mathbf{T}(\mathbf{k})$ are congruence transforms whose Bloch reduction $\mathbf{T}(\mathbf{k})$ is *independent of the design*, so the band sensitivity is once more a single $\partial\mathbf{D}_f/\partial\theta$ form, $\partial\lambda_n/\partial\theta_\ell = \boldsymbol{\psi}_n^{\mathsf{H}}(\partial\mathbf{K}/\partial\theta_\ell)\boldsymbol{\psi}_n$ with $\boldsymbol{\psi}_n = \mathbf{T}\boldsymbol{\phi}_n$, sampled over the Brillouin zone and needing no eigenvector derivative. Designing θ to suppress a chosen branch's dispersion (its spread across the zone) while a p-norm holds the neighbouring branches away flattens that band by $4.7\times$ (group velocity from 2.78 to 0.59) and opens complete gaps of $+0.42$ and $+0.66$ above and below it (Figure 11) — a slow-wave, mode-isolating cell built by fiber steering alone, printability governed by the same $|\zeta| \le \zeta_{\mathrm{all}}$. The band-structure design thus lives on a manufacturability front of the same kind, closing the argument that the front is a property of steered-fiber design in general, not of any one objective.

6.8. Composing the regimes: a multifunctional panel

The regimes above are usually studied one at a time; a real part must do several at once. We close with a single bolted CFRP mounting panel whose *one* orientation field is asked to do four jobs simultaneously, over a frequency *band*: deliver an incident machine vibration to an energy-harvesting sensor patch S, hold a mounted payload P in an acoustic shadow, keep the four fatigue-critical fastener rims F quiet, and — for

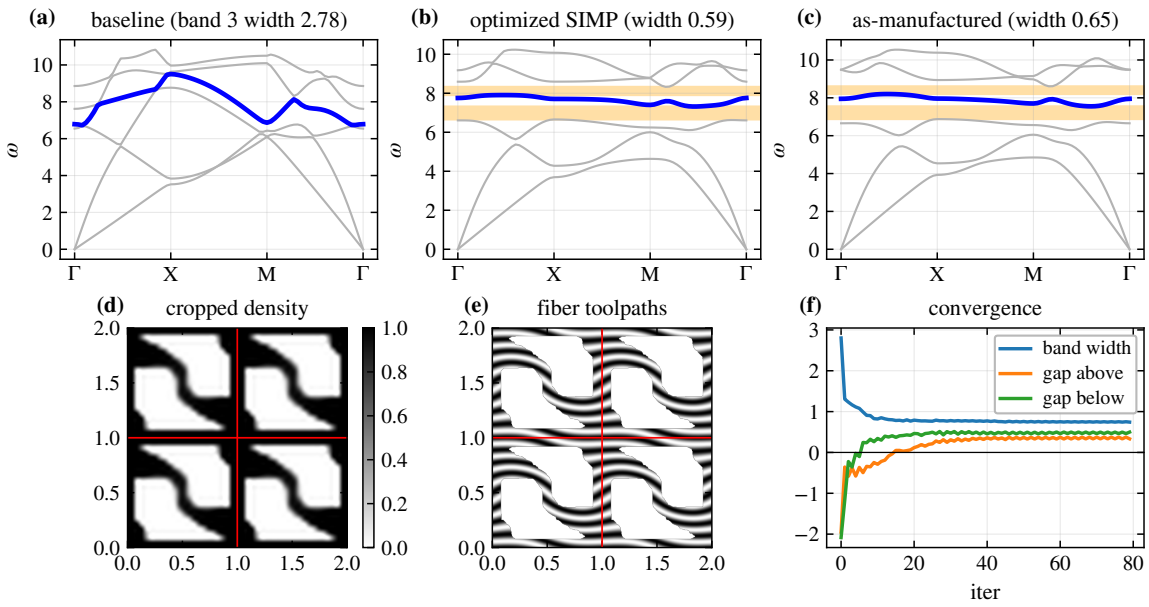


Figure 11: **The periodic front.** Fiber orientation on a Bloch cell flattens a target band by $4.7\times$ and opens complete gaps ($+0.42$, $+0.66$) on both sides, isolating a slow, nearly non-dispersive branch — all through the orientation field, under the same fiber-curvature limit.

free — stiffen the bolted mount. The wave jobs compose into one signed weight over the region energies $E_R = \int_{\Omega_R} |\mathbf{u}|^2$,

$$\mathcal{J}_{\text{wave}} = W_S \log \frac{E_S}{E_S^0} - \lambda_P \frac{E_P}{E_P^0} - \lambda_F \frac{E_F}{E_F^0} - \lambda_O \frac{E_O}{E_O^0}, \tag{18}$$

each term normalised to its straight-fiber baseline E_R^0 (all $O(1)$ and comparable), with O the interior outside the target regions (a leakage/confinement term) and the delivery reward *logarithmic* so protecting the rims stays worthwhile however bright the sensor already is. The gradient is a single adjoint per solve: for $\mathcal{J}_{\text{wave}} = \sum_R c_R E_R$ the adjoint load is $\sum_R (\partial \mathcal{J}/\partial E_R)\, w_R$, and (6) then returns $\mathrm{d}\mathcal{J}/\mathrm{d}\theta_\ell$. Robustness over the operational band $[44, 56]$ is imposed by averaging over 5 sample frequencies, $\mathcal{J} = \frac{1}{K}\sum_k \mathcal{J}_{\text{wave}}(\omega_k)$ — one adjoint per frequency.

The same $\hat{\boldsymbol{\theta}}$ is scored for stiffness: the panel is clamped at the four bolt rims and carries a service load over the payload footprint, and the compliance $c = \mathbf{f}^{\mathsf{T}}\mathbf{u}$ with its self-adjoint sensitivity (§3) enters as a co-objective. Because the fiber angle is largely spent on the wave, orientation buys only a modest stiffness co-benefit — $1.15\times$ over straight fibers, at no added mass — but it is genuinely *free*, riding along with the wave design.

The optimized layup (Figure 12) splits the incident beam cleanly around the payload into a dark pocket (shielded to $0.20\times$ its straight-fiber energy), threads the quiet gaps between the fastener rows (rims at $0.14\times$), and reconverges on the harvester ($1.7\times$ delivery), while lifting the mount stiffness $1.15\times$ — all in one printable, mirror-symmetric fiber field ($\max|\zeta| = 6.1 \le \zeta_{\text{all}}$) that holds across the band. Composition is the point: the manufacturability front is not four separate tricks but one design lever exercised against four objectives at once.

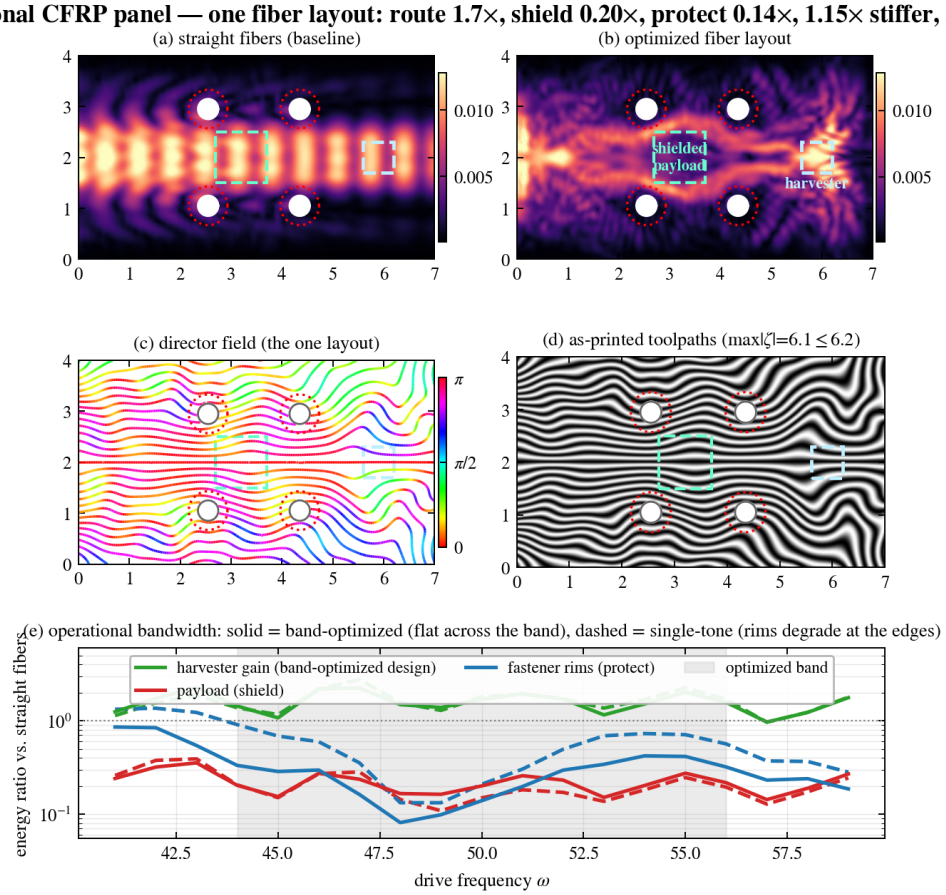


Figure 12: **A multifunctional CFRP panel.** One mirror-symmetric, curvature-feasible fiber layout routes an incident vibration to a harvester ($1.7\times$), shields a payload core ($0.20\times$), protects the four fastener rims ($0.14\times$), and stiffens the bolted mount ($1.15\times$, orientation only) — held across the operational band $[44, 56]$. (a) straight-fiber baseline; (b) optimized field, shielded payload (green) and harvester (blue) marked; (c) the director field; (d) the as-printed toolpaths; (e) the operational-bandwidth curve (band-optimized solid, single-tone dashed).

7. Discussion

The front is the result.. Across statics, waves, and periodic media the deliverable is not a single optimum but a front of performance versus the tow curvature a process allows. This reframes "manufacturability" from a yes/no gate into a design axis shared by every regime, because all three are driven by the same $\partial \mathbf{D}_f / \partial \theta$. A capable AFP head that lays tight radii buys more stiffness, more cloaking, a flatter band; a coarser process lands lower on the same curve.

Cloaking is the extreme point.. Elastic cloaking is where the front's structure is starkest: its ideal endpoint is not just unmanufacturable but *singular* (Norris [2]), so no relaxation of the process ever reaches it, and imposing the ideal's directions on a real material barely helps. That is exactly why direct optimization — which never tries to realize the singularity — is not a convenience but the only route to a manufacturable cloak, and it beats the analytical prescription roughly five to one while remaining printable.

Why fibers, and why bounded.. Unlike a lattice or a resonator array, the optimized medium here is a *solid, load-bearing* plate whose single design field is the fiber angle, laid directly by AFP. Its bounded contrast is what makes the transformation ideal unreachable, and its local orientability is what lets optimization approach the front from the manufacturable side.

Limitations.. The study is two-dimensional and single-frequency per solve, and no specimen has been fabricated; the natural next steps are broadband objectives, a 3D shell extension, and an AFP or 3D-printed coupon to close the loop on printability. The curvature constraint is an adaptive penalty with a best-feasible safeguard rather than a hard constraint with guarantees; casting it as an augmented-Lagrangian constraint and the front as a rigorous ε-constraint Pareto would strengthen the formulation.

8. Conclusion

We argued that a single *manufacturability front* — achievable performance versus allowed tow curvature — organizes fiber-orientation design of solid carbon-fiber parts across static stiffness, time-harmonic wave control, and phononic band structure, because all three carry the design through the same rotated fiber tensor and its closed-form sensitivity. On that front a bracket reaches 57% of a pointwise stiffness ceiling while staying printable ($9.4\times$ stiffer than an equal-mass plate); a lens focuses $12\times$; a Bloch cell flattens a band $4.7\times$; and a cloak — whose transformation ideal is the singular, unreachable endpoint — is restored from a useless $1.9\times$ analytical prescription to a printable $9.6\times$ optimized one. The manufacturable design is a chosen point on the front; the front is the object. The formulation is verifiable throughout: a real four-field elastodynamic solve, one adjoint lever for three physics, an exact isotropy-annihilation certificate, and conforming voids with zero interior energy.

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